

SIMULATOR EXAMINATION SCENARIO GUIDE

SCENARIO TITLE: 14-01 NRC ESG-3
SCENARIO NUMBER: 14-01 NRC ESG-3
EFFECTIVE DATE: See approval Dates Below
EXPECTED DURATION: 90 minutes
REVISION NUMBER: 00

PROGRAM: ☐ L.O. REQUAL
☒ INITIAL LICENSE
☐ STA
☐ OTHER _____

Revision Summary

New issue for 14-01 ILOT NRC Exam

PREPARED BY: G Gauding
Lead Regulatory Exam Author

8-31-15
Date

APPROVED BY: *Pat Williams*
Operations Training Manager

10/7/15
Date

APPROVED BY: *MW (mvl)*
Facility Representative

10915
Date

SCAN OF SIGNED SCENARIO COVER SHEET

I. OBJECTIVES

- A. Given the order or indications of a loss or malfunction of a safety related plant cooling water system, perform actions as the nuclear control operator to RESPOND to the loss or malfunction of the safety related plant cooling water system in accordance with S1/S2.OP-AB.SW-0001.
- B. Given indication of a loss or malfunction of a safety related plant cooling water system, DIRECT the response to the loss or malfunction in accordance with the approved station procedures.
- C. Given the order or indications of a loss of vacuum, perform actions as the nuclear control operator to RESPOND to the malfunction, in accordance with S1/S2.OP-AB.COND-0001.
- D. Given the order or indications of a loss of vacuum, DIRECT the response to the malfunction in accordance with S1/S2.OP-AB.COND-0001.
- E. Given the order or indications of a feedwater or condensate system malfunction, perform actions as the nuclear control operator to RESPOND to the malfunction in accordance with S1/S2.OP-AB.CN-0001.
- F. Given indication of a feedwater or condensate system malfunction, DIRECT the response to the malfunction in accordance with S1/S2.OP-AB.CN-0001.
- G. Given the order or indications of a loss of secondary heat sink, perform actions as the nuclear control operator to RESPOND to the loss of heat sink in accordance with the approved station procedures.
- H. Given indication of a loss of secondary heat sink, DIRECT the response to the heat sink loss in accordance with the approved station procedures.

II. MAJOR EVENTS

- A. 25 SW pump discharge strainer clogging.
- B. Loss of Condenser vacuum → downpower.
- C. Stuck Control rod during downpower.
- D. Condensate pump trip → SGFP trip.
- E. 2B 4KV vital bus UV.
- F. Loss of all AFW with no centrifugal charging pumps avail.
- G. FRHS Bleed and Feed w/Condensate pump recovery.

III. SCENARIO SUMMARY

- A. The crew will take the watch 85% power, BOL. 22 Charging pump is C/T for biofouling. 26 SW pump is C/T for strainer cleaning. 23 Condensate pump C/T. Power is reduced due to 23 condensate pump trip yesterday.

- B. Shortly after assuming the watch, 25 Service Water pump strainer will develop a high D/P. The crew will respond to alarms and board indications and dispatch an operator to investigate. The CRS will determine 25 SW pump is inoperable based on high D/P reported from field. The CRS will enter LCO 3.7.4 based on not having at least 1 SW pump operable on "C" vital bus.
- C. After the SW pump malfunction is addressed, 24 Main Condenser Vacuum pump will trip, and its discharge valve will not auto close. The CRS will enter S2.OP-AB.COND-0001, Loss of Main Condenser Vacuum as condenser vacuum slowly lowers. None of the standby vacuum pumps will start. The crew will perform a load reduction in an attempt to stabilize condenser vacuum.
- D. During the power reduction, the crew will identify a single Control Bank D rod not moving as rod insertion occurs. The CRS will enter S2.OP-AB.ROD-0001, Immovable / Misaligned Control Rods. The RO will place control rods in manual, and the downpower will be continued using boration only.
- E. Once the stuck rod has been addressed, one of the two inservice Condensate pumps will trip. SGFP suction pressure will rapidly and irreversibly lower to the point where both SGFPs will trip on low suction pressure. The crew will identify no SGFP's operating > P-10 (10% power), and initiate a manual reactor trip.
- F. The RO will perform EOP-TRIP-1, Reactor Trip or Safety Injection, Immediate Actions, and the CRS will transition to EOP-TRIP-2, Reactor Trip Response, with no SI required.
- G. When the Main Generator output breakers open 30 seconds after the Turbine trip, 2B 4KV vital bus will become deenergized. 2B EDG will start and load Blackout loads. 21 Charging pump and 22 Aux Feed pump fail to start on SEC initiation. After resetting 2B SEC, 21 Charging pump will trip after starting, and 22 AFW pump will not start.
- H. 21 AFW pump will trip 4 minutes after the Rx trip, leaving 23 AFW pump as the only operating aux feed pump. When SG NR levels have recovered but still remain <9%, 23 AFW pump will overspeed and trip.
- I. With CFST's in effect after transitioning out of TRIP-1, a RED path will be present for Heat Sink, and the CRS will transition to FRHS-1, Response to Loss of Secondary Heat Sink. The crew will identify no centrifugal charging pumps available and go to Bleed and Feed steps to stop RCPs, initiate SI, and open both PZR PORVs. (CT#1)
- J. The crew will continue to depressurize a SG to establish Condensate System flow to the SG. Once the crew has performed the steps to determine the feed rate, the Lead Evaluator will cue the crew that the selected SG NR level has reached 9%. The crew will then take actions to close the open PZR PORVs. (CT#2)
- K. The scenario will be terminated after both PZR PORVs are shut.

IV. INITIAL CONDITIONS

___ IC-233

PREP FOR TRAINING (i.e. computer setpoints, procedures, bezel covers, tagged equipment)

Initial	Description
___ 1	VC1and VC4 C/T
___ 2	RCPs (SELF CHECK)
___ 3	RTBs (SELF CHECK)
___ 4	MS167s (SELF CHECK)
___ 5	500 KV SWYD (SELF CHECK)
___ 6	SGFP Trip (SELF CHECK)
___ 7	23 CV PP (SELF CHECK)
___ 8	22 Charging pump C/T
___ 9	26 SW pump C/T
___ 10	23 Condensate pump C/T
___ 11	Complete Attachment 2 "Simulator Ready-for-Training/Examination Checklist."

Note: Tables with blue headings may be populated by external program, do not change column name without consulting Simulator Support group

EVENT TRIGGERS:

Initial	ET #	Description
	1	EVENT ACTION: kck02pbz // 22 VACUUM PUMP-START COMMAND: PURPOSE: <update as needed>
	3	EVENT ACTION: kck05pbz // 25 VACUUM PUMP-START COMMAND: PURPOSE: <update as needed>
	5	EVENT ACTION: MONP254 <10 // CONT ROD BANK C < 10 (RX TRIP) COMMAND: PURPOSE: <update as needed>
	7	EVENT ACTION: kb417pbj // 21 CHG PUMP-START COMMAND: PURPOSE: <update as needed>
	9	EVENT ACTION: MONP254 <10 // CONT ROD BANK C < 10 (RX TRIP) COMMAND: PURPOSE: <update as needed> DMF CN0086B
	11	EVENT ACTION: MONP254 <10 // CONT ROD BANK C < 10 (RX TRIP) COMMAND: PURPOSE: <update as needed>DMF VL0037

MALFUNCTIONS:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Severity
01	SW0222E 25 SW PUMP DISCH Clogging	N/A	5	00:07:00	RT-1	26
02	VC0087C TRIP OF VACUUM PUMP 24	N/A	N/A	N/A	RT-3	
03	VL0337 24AR25 Fails to Position (0-100%)	N/A	N/A	N/A	N/A	100
04	VC0087A TRIP OF VACUUM PUMP 22	00:00:02	N/A	N/A	ET-1	
05	CN0086B Loss of 22 main condenser vacuum	00:00:10	5	00:10:00	ET-3	7
06	RD0065 ANY ROD(S) FAILS TO MOVE (ELECT	N/A	N/A	N/A	N/A	50
07	CN0117B 22 CONDENSATE PUMP TRIP	N/A	N/A	N/A	RT-5	
08	AF0353B 22 AFP FAILURE TO AUTO START ON ANY (ALL) SIGNALS	N/A	N/A	N/A	N/A	
09	RP318S1 21 CHRG PMP FAILS TO STRT ON SEC	N/A	N/A	N/A	N/A	
10	CV0208A 21 CHARGING PUMP TRIP	00:00:02	N/A	N/A	ET-7	
11	RP318E2 Aux Feed Pump 22 Fails to Start on SEC	N/A	N/A	N/A	N/A	
12	AF0181A 21 AUX FEEDWATER PUMP TRIP	00:04:00	N/A	N/A	ET-5	

13	AF0183 23 AUX FW PMP OVERSPEED TRIP	N/A	N/A	N/A	RT-9	
14	CV0034 23 CHARGING PUMP TRIP	00:00:01	N/A	N/A	RT-14	

REMOTES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition
01	RC05A RCS SYSTEM , BORON CONC RESET	N/A	N/A	N/A	N/A	892
02	SW43D 26 SW PUMP BKR CONTROL POWER	N/A	N/A	N/A	N/A	OFF
03	SW44D 26 SW PUMP RACK OUT	N/A	N/A	N/A	N/A	TAGGED
04	CN38D 23 CN PUMP BKR CONTROL POWER	N/A	N/A	N/A	N/A	OFF
05	CN39D 23 CN PUMP BKR RACK OUT	N/A	N/A	N/A	N/A	TAGGED
06	CV48D 22 CHG PUMP BKR CONTROL POWER	N/A	N/A	N/A	N/A	OFF
07	CV49D 22 CHG PUMP RACK OUT	N/A	N/A	N/A	N/A	TAGGED
08	SP01A1 Prevailing Wind Speed	N/A	N/A	N/A	N/A	10
09	SP01A2 Prevailing Wind Direction	N/A	N/A	N/A	N/A	160

OVERRIDES:

SELF-CHECK	Description	Delay Time	Initial Value	Ramp Time	Trigger	Condition/Severity
01	C808 F DI 24BSD 2B VITAL BUS FEEDER-OPEN	00:00:30	N/A	N/A	ET-5	ON
02	C805 F DI 23BSD 2B VITAL BUS FEEDER-OPEN	00:00:30	N/A	N/A	ET-5	ON
03	B512 E DI 22 AUX FEED PUMP-START	N/A	N/A	N/A	N/A	OFF
04	B111 2CC71 LETDOWN TEMP-MANUAL	N/A	N/A	N/A	RT-14	ON

OTHER CONDITIONS:

Description
1. None

V. SEQUENCE OF EVENTS

- A. State shift job assignments.
- B. Hold a shift briefing, detailing instruction to the shift: (provide crew members a copy of the shift turnover sheet).
- C. Inform the crew "The simulator is running. You may commence panel walkdowns at this time. CRS please inform me when your crew is ready to assume the shift".
- D. Allow sufficient time for panel walk-downs. When informed by the CRS that the crew is ready to assume the shift, ensure the simulator is cleared of unauthorized personnel.

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
1. 25 SW pump discharge strainer high D/P			
Simulator Operator: Insert <u>RT-1</u> on direction from Lead Evaluator. MALF: SW0222E 25 SW PUMP DISCH Clogging Initial Value: 5 Final Value: 26 Ramp: 7 minutes			
Note: It will take 01:40 minutes for the first alarm to annunciate. Role Play: 2 minutes after being dispatched, report that 25 SW pump strainer is in auto with backwash red light on, you can hear flow through the 25SW24 backwash valve, and the strainer shaft is NOT turning with the strainer motor running. 25 SW pump strainer D/P is ~ 10.5 psid			
	RO either notices lowering SW header pressures, or reports OHA B-16, 24-26 SW STRAIN TRBL as unexpected when it occurs.		
	CRS dispatches an operator to investigate alarm.		
	CRS directs local operator to collect Trip/High		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: The Attachment 2 is found at the end of this scenario guide.	D/P Observation IAW S2.OP-SO.SW-0001, Service Water Pump Operation.		
	CRS may direct swapping to an operable SW pump after Trip/High D/P data collection.		
	CRS determines manual backwash cannot be performed with strainer not turning.		
	CRS determines 25 SW pump is inoperable based on higher than 10 psid across strainer IAW S2.OP-AR.ZZ-0002, Overhead Annunciators Window B.		
	CRS contacts Maintenance to assist in 26 SW pump strainer repair.		
Note: If contacted about status of work on <u>26</u> SW pump, report that Maintenance Supervisor has just signed off tags.			
	CRS enters LCO 3.7.4 based on not having any operable SW pumps powered from "C" vital bus, causing only ONE SW loop to be operable.		
Continue to next event after Tech Spec determination on direction from Lead Evaluator.			
2. Loss of Main Condenser			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Vacuum			
Simulator Operator: Insert RT-3 on direction from Lead Evaluator. MALFS: VC0087C TRIP OF VACUUM PUMP 24 VL0337 24AR25 Fails to Position (100%)			
	PO reports trip of 24 Vacuum pump, and 24AR24 has not automatically shut.		
	CRS directs PO to shut 24AR25 (either here or when directed in AB.COND)		
	PO reports 24AR25 will not shut, and condenser backpressure is slowly rising.		
	CRS enters S2.OP-AB.COND-0001, Loss of Condenser Vacuum.		
	CRS directs initiation of Attachment 1 CAS.		
	CRS dispatches operators to perform Attachment 2 Local Vacuum checks.		
Role Play: 5 minutes after being dispatched, report local vacuum pump checks complete, with only 24AR25 being open with 24 vacuum pump secured as abnormal.			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	If not performed earlier, CRS directs closure of 24AR25.		
	CRS determines vacuum problem is not Circ Water related.		
	CRS directs PO to start all available vacuum pumps.		
Simulator Operator: Ensure ET-1 and ET-3 are true when 22 and 25 vacuum pump start PBs are depressed respectively. This causes 22 vacuum pump to trip 2 seconds after starting, and inserts the loss of vacuum malfunction.			
Simulator Operator: Modify MALF CN0086B if necessary to ensure condenser vacuum continues to slowly degrade, but does not reach any required trip setpoints.			
	PO reports condenser backpressure cannot be maintained stable or lowering.		
	PO reports the Turbine is latched.		
	CRS directs the RO and PO to coordinate and perform a power reduction to attempt to stabilize condenser backpressure.		
Note: CRS direction should include a target point for the load reduction,			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
and the rate at which the power reduction is to be performed. Note: If AB.LOAD is entered the following steps will be performed. Note: No further AB.LOAD actions will be required. AB.COND steps resume here.			
	CRS may enter S2.OP-AB.LOAD-0001, Rapid Load Reduction to perform the load reduction.		
	CRS directs initiation of AB.LOAD CAS.		
	PO initiates a turbine load reduction to target point at the rate which CRS directed the load reduction be performed.		
	RO maintains Tavg on program using control rods and boration.		
	RO energizes PZR heaters.		
	RO calculates boron addition and expected rod position for power reduction directed by CRS, and recommends a rate at which to perform a boration.		
	CRS ensures boration calculation is correct, and RO initiates boration at rate directed by CRS.		
	PO monitors condensate pump suction temps,		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	and removes SGBD from service by closing 21-24GB4's and 21-24GB185s if they reach 120°F, and bypasses the Polisher if they reach 140°F.		
	RO reports when expected rod movement occurs.		
	PO reports condenser backpressure continues to slowly degrade.		
	Crew ensures Attachment 4 Condenser Backpressure Limits are monitored and not exceeded.		
	RO reports when stuck rod 2D2 is suspected.		
	When stuck rod is verified to not be moving as expected, CRS centers S2.OP-AB.ROD-0001, Immovable / Misaligned Control Rods.		
	RO places Rod Bank Selector Switch in manual.		
	CRS determines the load reduction must be continued, and directs RO to raise the boration rate, and the PO to lower turbine load reduction rate if necessary to maintain Tavg within +/- 1.5°F of program.		
Simulator Operator: IF the CRS briefs that a Rx trip will be performed based on degrading plant conditions, THEN RT-5 should be inserted to trip 22			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Condensate pump prior to that direction being given.			
Note: Depending on when the rod misalignment is detected, the control rod may or may not be >18 steps mis-aligned with Rx power <85%. Actions performed will be the same regardless of the amount of misalignment. Note: If the power reduction continues far enough, OHA E-38 UPPER SECT DEV ABV 50% PWR will alarm. Role Play: When contacted, report BEACON indicates rod misalignment is confirmed.	RO reports OHA E-40 ROD BANK URGENT FAIL is clear.		
	PO reports rod misalignment is indicated on plant Computer.		
	RO monitors QPTR and AFD.		
	CRS contacts Rx Engineering to confirm misalignment.		
	RO reports only a single rod is misaligned.		
	CRS enters LCO 3.1.3.1 for one misaligned rod.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Continue to next event on direction from Lead Evaluator.	If power reduction has been stopped >75% power, CRS directs power reduction to <75% RTP to comply with action of TSAS 3.1.3.1 Action c.3.d.		
Simulator Operator: Insert RT-5 on direction from Lead Evaluator: MALF: CN0117B 22 CONDENSATE PUMP TRIP			
Note: SGFP trip will occur within a few seconds on low suction pressure. Note: The Rx will automatically trip on lo lo SG NR level if a manual Rx trip is not performed.	PO reports 22 condensate pump trip, and rapidly lowering SGFP suction pressure.		
	PO may open 21-23CN108 Condensate Polisher Bypass valves, and reports SGFP suction pressure continues to degrade.		
	PO reports both SGFPs have tripped.		
	RO or PO trips the Rx based on no SGFPs operating with Rx power >P-10 (10% power)		
Simulator Operator: Ensure ET-5			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
is true upon the Rx trip. This starts a 4 minute timer to trip 21 AFW pump.			
	RO performs Immediate Actions of TRIP-1 - Confirms the Rx trip - Trips the Main Turbine - Reports at least one 4KV Vital bus energized. - Reports SI has not actuated, nor is it required.		
Simulator Operator: Ensure ET-9 and ET-11 are true upon the Rx trip. This deletes the vacuum malfunctions.			
	CRS and RO verify performance of immediate actions.		
	CRS transitions to EOP-TRIP-2 Reactor Trip Response.		
	PO reports 2B 4KV vital bus has loaded in Blackout mode.		
Note: Direction for responding to Blackout Loading occurring on any vital bus is at Step 12 of TRIP-2.			
	PO reports 22 AFW pump has not started.		
	PO receives permission to throttle AFW flow to no less than 22E4 lbm/hr.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: With CFSTs in effect after transition out of TRIP-1, AFW flow <22E4 lbm/hr with SG NR levels <9% is entry criteria for FRHS-1 on RED path. Entry may not be performed if condition is not present after verifying no higher RED paths.	RO makes page announcement.		
	CRS directs implementation of ECG.		
	PO reports total AFW flow is >22E4 lbm/hr and both SGFPs are tripped.		
	PO maintains at least 22E4 lbm/hr AFW flow.		
	RO reports RCPs are running and Tavg trending to or at 547°F.		
	RO reports both RTBs are open.		
	With Tavg <554°F, PO ensures 21-24BF19, 21-24BF40, and 21-24BF22 are shut.		
	RO reports control rods inserted on Rx trip.		
	PO reports when 21 AFW pump trips, and adjusts 23 AFW pump speed as necessary to ensure 22E4 lbm/hr is maintained.		
	RO reports PZR level is >17%, charging flow is established, and maintains PZR level 22%.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: Blocking 2B SEC on 2RP1 is not directed since there is no SI signal present.	RO reports no fire on Unit 1, and letdown flow is established.		
	RO reports PZR pressure stable or trending to 2235 psig.		
	PO maintains AFW flow >22E4 lbm/hr until at least one SG NR level is >9%, then maintains intact SG NR levels between 9-33%.		
	PO reports all 4KV Group buses energized.		
	PO reports Blackout loading has occurred for 2B vital bus.		
	PO reports Blackout loading is not complete, and available equipment has failed to start.		
	PO resets 2B SEC.		
Simulator Operator: IF CRS does NOT start 21 charging pump, <u>then</u> trip 23 charging pump here by inserting <u>RT-14</u>.	PO reports 21 charging pump started then tripped.		
Simulator Operator: Insert <u>RT-9</u> to trip 23 AFW pump after 21 charging pump start has been			

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
attempted.			
	PO reports 22 AFW pump will not start.		
	PO reports when 23 AFW pump trips.		
	Crew determines a RED path is present when AFW flow is <22E4 lbm/hr and ALL SG NR levels are <9%.		
	CRS transitions to FRHS-1, Response to Loss of Secondary Heat Sink.		
	PO reports loss of all AFW was not due to operator action.		
	RO reports RCS pressure is > all intact or ruptured SG pressures, and RCS Thots are >350°F.		
	RO reports neither 21 nor 22 charging pumps are available.		
	CRS goes to Step 23 Bleed and Feed initiation.		
	RO stops all RCPs.		
	RO initiates SI.		
	PO reports SI valves in safeguards positions.		
	RO reports no centrifugal charging pumps running, and both SI pumps running.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
CT#1 (CT-46) Initiate RCS bleed and feed so that the RCS depressurizes sufficiently for SI pump injection into RCS. SAT UNSAT	RO reports correct valve alignment for both SI pumps.		
	RO reports both PZR PORV Stop valves are open, and opens BOTH PZR PORVs.		
	RO reports both PZR PORV Stop valves and both PZR PORVs are open.		
	PO reports SI Valve alignment (prior to SI reset using 2RP4 indication) or performs APPX-3 SI Verification.		
	RO resets SI and Phase A isolation, and reports Phase B isolation reset.		
	RO opens 21 and 22CA330.		
	PO resets all SECs, and resets 230V control centers as required.		
	Crew maintains ECCS flow and PZR PORVs open.		
	RO reports containment pressure < 15 psig, and no containment spray pumps running.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: SGFPs are not available based on no SW flow to TGA, and all AFW pumps have tripped.	CRS determines only success path is Condensate pump recovery, and goes to Step 20.		
	CRS determines 22 or 24 SG should be depressurized, whichever has the LOWEST level.		
	PO reports condenser steam dumps are available.		
	RO initiates MSLI for all loops except selected SG.		
	RO places steam dumps in manual, adjusts valve demand to 0%, then places in MS Pressure Control.		
	RO adjusts steam pressure valve demand to 25%, and bypasses Tavg at 543°F.		
Simulator Operator: 2 minutes after being dispatched, insert RT-11 or RT-13 to open 22BF40 or 24BF40 as directed.	CRS dispatches an operator to open selected SG BF40 or BF19.		
	PO ensures selected SG BF13 is open.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
Note: When step 20.5 is reached "Is Condensate Flow Established to Any SG", the CRS should wait to see if actions taken to establish condensate flow are successful when the selected SG depressurized sufficiently to the point where condensate flow should occur. IF CRS prematurely answers "no" at step 20.5, they will go back to step 3 and needlessly verify all the Bleed and Feed steps.	PO ensures opens selected SG BF22.		
	PO opens 21 and 22CN48s (SGFP bypass), and shuts 21 and 22 CN32s (SGFP suct valve)		
	PO reports when condensate flow is indicated to selected SG.		
	PO reports no SG NR levels are >9%, and WR level is rising in selected SG.		
	PO maintains selected SG pressure <575 psig.		
	CRS determines Bleed and Feed has been initiated and goes to Step 36.		
	RO reports CET trends.		
	IF CETs are lowering, CRS directs feeding of selected SG at desired rate unless selected		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	SG WR level is <11%, when it should only be fed at 1-5E4 lbm/hr.		
	IF CETs are still rising, CRS directs feeding of selected SG at maximum rate.		
	PO reports selected SG is not faulted or ruptured.		
	PO reports other intact SG WR levels are >11%.		
Lead Evaluator: At your discretion, either advise the CRS directly or announce to the entire crew that time compression is being used, and that the selected SG NR level has risen to >9%.	PO reports no SG NR levels are >9%.		
	CRS returns to Step 35.		
	RO reports CET trends.		
	IF CETs are lowering, CRS directs feeding of selected SG at desired rate unless selected SG WR level is <11%, when it should only be fed at 1-5E4 lbm/hr.		
	IF CETs are still rising, CRS directs feeding of selected SG at maximum rate.		
	PO reports selected SG is not faulted or ruptured.		

Evaluator/Instructor Activity	Expected Plant/Student Response	SBT LOG	Comment
	PO reports other intact SG WR levels are >11%.		
	PO reports selected SG NR level is >9%.		
	RO reports CETs and RCS Thots are dropping.		
	RO reports Head Vent valves are shut.		
	RO reports subcooling is <50°F.		
	RO shuts ONE PORV.		
	RO reports charging pump suction aligned to the RWST.		
CT#2 (CT-47) Close all PZR PORVs and reactor vessel head vent valves before the end of the scenario SAT UNSAT	RO reports 2CV139 and 2CV140 are open.		
	RO shuts open PORVs.		
Terminate scenario when both PORVs are shut.			

VI. SCENARIO REFERENCES

- A. Alarm Response Procedures (Various)
- B. Technical Specifications
- C. Emergency Plan (ECG)
- D. OP-AA-101-111-1003, Use of Procedures
- E. S2.OP-SO.SW-0001, Service Water Pump Operation
- F. S2.OP-SO.SW-0005, Service Water System Operation
- G. S2.OP-AB.COND-0001, Loss of Main Condenser Vacuum
- H. S2.OP-AB.LOAD-0001, Rapid Load Reduction
- I. S2.OP-AB.ROD-0001, Immovable / Misaligned Control Rod
- J. S2.OP-AB.CN-0001, Main Feedwater / Condensate System Abnormality
- K. 2-EOP-TRIP-1, Reactor Trip or Safety Injection
- L. 2-EOP-TRIP-2, Reactor Trip Response
- M. 2-EOP-FRHS-1, Response to Loss of Secondary Heat Sink

ATTACHMENT 1
UNIT TWO PLANT STATUS
TODAY

MODE: 1 POWER: 85% RCS BORON: 893 MWe 1010

SHUTDOWN SAFETY SYSTEM STATUS (5, 6 & DEFUELED):

NA

REACTIVITY PARAMETERS

Control Bank D at 178 steps withdrawn

Xenon building in at 12 ppm per hour

200 gallon dilution for temperature control was completed 30 minutes ago.

MOST LIMITING LCO AND DATE/TIME OF EXPIRATION:

3.5.2.a action a for 22 charging pump C/T. Expires in 70 hours.

EVOLUTIONS/PROCEDURES/SURVEILLANCES IN PROGRESS:

None

ABNORMAL PLANT CONFIGURATIONS:

Power reduced to 85% IAW S2.OP-AB.CN-0001 when 23 Condensate pump tripped last shift.

CONTROL ROOM:

Unit 1 and Hope Creek at 100% power.

No penalty minutes in the last 24 hrs.

PRIMARY:

22 Charging pump C/T for oil cooler cleaning.

SECONDARY:

23 Condensate pump C/T, tripped last shift.

RADWASTE:

No discharges in progress

CIRCULATING WATER/SERVICE WATER:

26 SW pump C/T for strainer cleaning.

ATTACHMENT 2**SIMULATOR READY FOR TRAINING CHECKLIST**

- ___ 1. Verify simulator is in "TRAIN" Load
- ___ 2. Simulator is in RUN
- ___ 3. Overhead Annunciator Horns ON
- ___ 4. All required computer terminals in operation
- ___ 5. Simulator clocks synchronized
- ___ 6. All tagged equipment properly secured and documented
- ___ 7. TSAS Status Board up-to-date
- ___ 8. Shift manning sheet available
- ___ 9. Procedures in progress open and signed-off to proper step
- ___ 10. All OHA lamps operating (OHA Test) and burned out lamps replaced
- ___ 11. Required chart recorders advanced and ON (proper paper installed)
- ___ 12. All printers have adequate paper AND functional ribbon
- ___ 13. Required procedures clean
- ___ 14. Multiple color procedure pens available
- ___ 15. Required keys available
- ___ 16. Simulator cleared of unauthorized material/personnel
- ___ 17. All charts advanced to clean traces and chart recorders are on.
- ___ 18. Rod step counters correct (channel check) and reset as necessary
- ___ 19. Exam security set for simulator
- ___ 20. Ensure a current RCS Leak Rate Worksheet is placed by Aux Alarm Typewriter
With Baseline Data filled out
- ___ 21. Shift logs available if required
- ___ 22. Recording Media available (if applicable)
- ___ 23. Ensure ECG classification is correct
- ___ 24. Reference verification performed with required documents available
- ___ 25. Verify phones disconnected from plant after drill.
- ___ 26. Verify EGC paperwork is marked "Training Use Only" and is current revision.
- ___ 27. Ensure sufficient copies of ECG paperwork are available.

ATTACHMENT 3**CRITICAL TASK METHODOLOGY**

In reviewing each proposed CT, the examination team assesses the task to ensure, that it is essential to safety. A task is essential to safety if, in the judgment of the examination team, the improper performance or omission of this task by a licensee will result in direct adverse consequences or in significant degradation in the mitigative capability of the plant.

The examination team determines if an automatically actuated plant system would have been required to mitigate the consequences of an individual's incorrect performance. If incorrect performance of a task by an individual necessitates the crew taking compensatory action that would complicate the event mitigation strategy, the task is safety significant.

- I. Examples of CTs involving essential safety actions include those for which operation or correct performance prevents...
 - degradation of any barrier to fission product release
 - degraded emergency core cooling system (ECCS) or emergency power capacity
 - a violation of a safety limit
 - a violation of the facility license condition
 - incorrect reactivity control (such as failure to initiate Emergency Boration or Standby Liquid Control, or manually insert control rods)
 - a significant reduction of safety margin beyond that irreparably introduced by the scenario
- II. Examples of CTs involving essential safety actions include those for which a crew demonstrates the ability to...
 - effectively direct or manipulate engineered safety feature (ESF) controls that would prevent any condition described in the previous paragraph.
 - recognize a failure or an incorrect automatic actuation of an ESF system or component.
 - take one or more actions that would prevent a challenge to plant safety.
 - prevent inappropriate actions that create a challenge to plant safety (such as an unintentional Reactor Protection System (RPS) or ESF actuation).

ATTACHMENT 4

SIMULATOR SCENARIO REVIEW CHECKLIST

SCENARIO IDENTIFIER: 14-01 ILOT NRC ESG-3 REVIEWER: B Blose

Initials	Qualitative Attributes
BB	1. The scenario has clearly stated objectives in the scenario.
BB	2. The initial conditions are realistic, in that some equipment and/or instrumentation may be out of service, but it does not cue crew into expected events.
BB	3. The scenario consists mostly of related events.
BB	4. Each event description consists of: • the point in the scenario when it is to be initiated • the malfunction(s) that are entered to initiate the event • the symptoms/cues that will be visible to the crew • the expected operator actions (by shift position) • the event termination point
BB	5. No more than one non-mechanistic failure (e.g., pipe break) is incorporated into the scenario without a credible preceding incident such as a seismic event.
BB	6. The events are valid with regard to physics and thermodynamics.
BB	7. Sequencing/timing of events is reasonable, and allows for the examination team to obtain complete evaluation results commensurate with the scenario objectives.
BB	8. The simulator modeling is not altered.
BB	9. All crew competencies can be evaluated.
BB	10. The scenario has been validated.
BB	11. If the sampling plan indicates that the scenario was used for training during the requalification cycle, evaluate the need to modify or replace the scenario.
BB	12. ESG-PSA Evaluation Form is completed for the scenario at the applicable facility.

ATTACHMENT 4
SIMULATOR SCENARIO REVIEW CHECKLIST

Initial		Quantitative Attributes (as per ES-301-4, and ES-301 Section D.5.d)	
GG	3	Malfunctions that occur after EOP entry: 1-2	
GG	3	Abnormal Events: 2-4	
GG	1	Major Transients: 1-2	
GG	2	EOPs entered/requiring substantive actions: 1-2	
GG	1	EOP contingencies requiring substantive actions: 0-2	
GG	2	Crew Critical Tasks: 2-3	
COMMENTS:			

ATTACHMENT 3
ESG CRITICAL TASKS

14-01 ILOT NRC ESG-3

CT#1 (CT-46) Initiate RCS bleed and feed so that the RCS depressurizes sufficiently for SI pump injection into RCS.

Bases: Failure to initiate RCS bleed and feed before the RCS saturates at a pressure above the shutoff head of the high-head ECCS pumps results in significant and sustained core uncover. If RCS bleed is initiated so that the RCS is depressurized below the shutoff head of the high-head ECCS pumps, then core uncover is prevented or minimized.

CT #2 (CT-47) Close all PRZR PORVs and reactor vessel head vent valves before the end of the scenario

Bases: Failure to close the RCS vent paths (such as head vents and PRZR PORVs) under the postulated plant conditions constitutes misoperation or incorrect crew performance that fails to prevent "degradation to any barrier to fission product release." In this case, the RCS fission-product barrier can be restored to full integrity only if the PRZR PORVs and the reactor vessel head vents are closed. Failure to close the RCS vent paths also results in the crew having to transition to LOCA-1 because of the continuing LOCA, instead of being able to transition to TRIP-3 for SI termination. Thus, failure to perform the critical task "necessitates the crew taking compensating action that would complicate the event mitigation strategy."

ATTACHMENT 6
ESG-PSA RELATIONSHIP EVALUATION

EVENTS LEADING TO CORE DAMAGE

<u>Y/N</u>	<u>Event</u>	<u>Y/N</u>	<u>Event</u>
<u>N</u>	TRANSIENTS with PCS Unavailable	<u>N</u>	Loss of Service Water
<u>N</u>	Steam Generator Tube Rupture	<u>N</u>	Loss of CCW
<u>N</u>	Loss of Offsite Power	<u>N</u>	Loss of Control Air
<u>N</u>	Loss of Switchgear and Pen Area Ventilation	<u>N</u>	Station Black Out
<u>N</u>	LOCA		

COMPONENT/TRAIN/SYSTEM UNAVAILABILITY THAT INCREASES CORE DAMAGE FREQUENCY

<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>	<u>Y/N</u>	<u>COMPONENT, SYSTEM, OR TRAIN</u>
<u>N</u>	Containment Sump Strainers	<u>N</u>	Gas Turbine
<u>N</u>	SSWS Valves to Turbine Generator Area	<u>N</u>	Any Diesel Generator
<u>N</u>	RHR Suction Line valves from Hot Leg	<u>Y</u>	Auxiliary Feed Pump
<u>N</u>	CVCS Letdown line Control and Isolation Valves	<u>N</u>	SBO Air Compressor

OPERATOR ACTIONS IMPORTANT IN PREVENTING CORE DAMAGE

<u>Y/N</u>	<u>OPERATOR ACTION</u>
<u>N</u>	Restore AC power during SBO
<u>N</u>	Connect to gas turbine
<u>N</u>	Trip Reactor and RCPs after loss of component cooling system
<u>N</u>	Re-align RHR system for re-circulation
<u>N</u>	Un-isolate the available CCW Heat Exchanger
<u>N</u>	Isolate the CVCS letdown path and transfer charging suction to RWST
<u>N</u>	Cooldown the RCS and depressurize the system
<u>N</u>	Isolate the affected Steam Generator that has the tube rupture(s)
<u>N</u>	Early depressurize the RCS
<u>Y</u>	Initiate feed and bleed

Complete this evaluation form for each ESG.

S2.OP-SO.SW-0001(Q)

ATTACHMENT 2
(Page 1 of 2)

SERVICE WATER STRAINER TRIP/HIGH ΔP OBSERVATIONS (as applicable)

STRAINER # _____ TIME (problem noted) _____ DATE _____

1.0 Was the Service Water Pump just started? YES/NO

◆ In service for how long? _____ TIME

2.0 What was the first indication of trouble? OHA/OTHER

Comments: _____

3.0 Is the strainer turning? YES/NO

4.0 RECORD the following data (attempt even if strainer stops):

◆ Strainer ΔP _____ ΔP

◆ Service Water Pump discharge pressure _____ PSIG

◆ Strainer inlet pressure _____ PSIG

◆ Strainer outlet pressure _____ PSIG

5.0 Was the pump discharge valve (SW3) closed to backflush the affected strainer? YES/NO

6.0 IF strainer overloads were found tripped and successfully reset,
THEN REQUEST Maintenance to take strainer AMP readings.

7.0 Is the associated traveling screen operating? YES/NO

◆ Is there any carryover at the traveling screen? YES/NO

◆ What is the traveling screen ΔP ? _____ ΔP

8.0 Is another strainer in the same Service Water Bay also in backwash? YES/NO

9.0 What position is the backwash valve in, based on stem? _____ OPEN/CLOSED/MID

S2.OP-SO.SW-0001(Q)

ATTACHMENT 2
(Page 2 of 2)

SERVICE WATER STRAINER TRIP/HIGH)P OBSERVATIONS (as applicable)

10.0 Is the sound of backflush clearly heard? YES/NO

11.0 Backwash flow readings (if avail from Maint). _____ GPM

12.0 Is flow observed in the blowdown sump? YES/NO
(outside left, prior to going up the stairs at the Service Water Bldg)

13.0 Additional Comments:
